

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTEPC602 - ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1. What is the main goal of Artificial Intelligence?
 - A) To create a database
 - B) To solve complex mathematical problems
 - C) To develop machines that can think and act like humans
 - D) To design faster computers

What is an AI agent?

 - A. A software that only collects data
 - B. A program that always requires human input
 - C. An entity that perceives its environment and acts upon it
 - D. A physical robot only
3. In a vacuum cleaner world, which of the following would be a rational agent's behaviour?
 - A. Move randomly
 - B. Move and clean based on percepts
 - C. Clean only one corner repeatedly
 - D. Ignore percepts and do nothing
4. The four main types of agents are:
 - A. Reactive, Heuristic, Rational, and Learning
 - B. Simple reflex, Model-based, Goal-based, and Utility-based
 - C. Static, Dynamic, Adaptive, and Heuristic
 - D. Passive, Active, Intelligent, and Emotional

What does 'rationality' in Artificial Intelligence typically refer to?

 - A) The ability to think like a human
 - B) The ability to always be correct
 - C) The ability to act in a way that maximises goal achievement
 - D) The ability to store and retrieve data efficiently
6. Which of the following algorithms uses only the heuristic function $h(\square)$ to guide the search?
 - A) A* Search
 - B) Uniform Cost Search
 - C) Depth-First Search
 - D) Greedy Best-First Search

- 7 Overfitting occurs when:
 A. The model performs well on new data
 B. The model generalises well
 C. The model performs well on training data but poorly on test data
 D. The model has too few parameters
- 8 The confusion matrix is used to evaluate:
 A. Regression models
 B. Classification models
 C. Clustering models
 D. Time series forecasting
- 9 What is the main strategy of the Hill Climbing algorithm?
 A) Explore all neighbours before choosing
 B) Always move to a neighbour with a lower cost
 C) Move to the neighbour with the highest heuristic value
 D) Move randomly to avoid local maxima
- 10 Which function is used in Logistic Regression to map values between 0 and 1?
 A. Linear function
 B. Exponential function
 C. Sigmoid function
 D. Hyperbolic tangent
- 11 Which of these is NOT a common evaluation metric for classification problems?
 A. Accuracy
 B. Precision
 C. Mean Squared Error
 D. Recall
- 12 Which of the following is the cost function used in Linear Regression?
 A) Cross-Entropy
 B) Hinge Loss
 C) Mean Squared Error
 D) Log Loss
- Q2. Solve the following.
- A) What are the stages of Artificial Intelligence? Explain with examples. 6
- B) What are the types of Agents? Explain any one. 6
- Q3. Solve the following.
- A) Differentiate between the depth-first search and breadth-first search Algorithms. 6
- B) What is TABU Search? Explain in detail. 6
- Q4. Solve Any Two of the following.
- A) What are the types of Machine learning? Explain in detail. 6
- B) What is the difference between Classification and Regression? Explain. 6
- C) What is a Decision tree? Elaborate with an example. 6
- Q5. Solve Any Two of the following.
- A) What is a confusion matrix? Explain. 6
- B) Explain the performance metrics for Classification problems in detail. 6
- C) Discuss the performance metrics for Regression problems in detail. 6
- Q6. Solve Any Two of the following.
- A) Discuss the types of linear regression models with the cost function. 6

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| B) | What is Gradient Descent? Discuss its variants in detail. | 6 |
| C) | Differentiate between Linear Regression and Logistic Regression | 6 |

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